Scipion-chem: an open platform for virtual drug screening

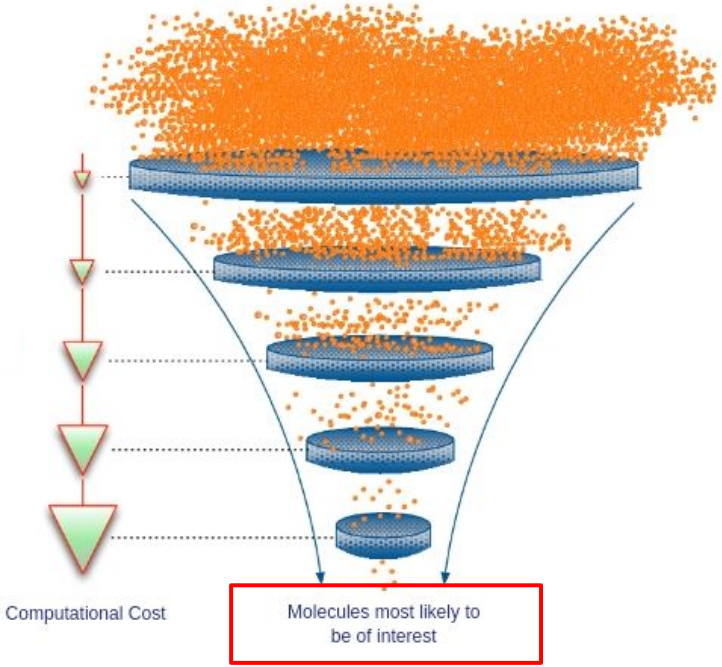
Selective docking on PI3K isoforms

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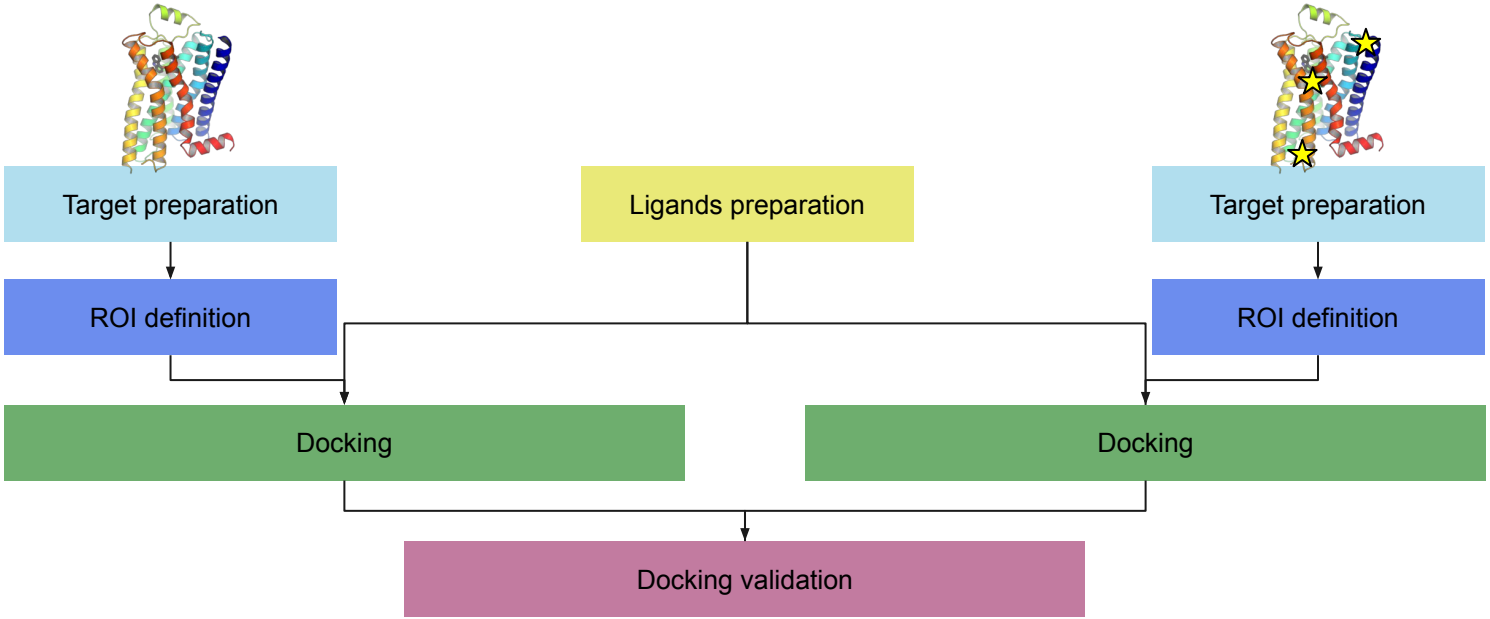
17/12/2025

Selective VDS: overview

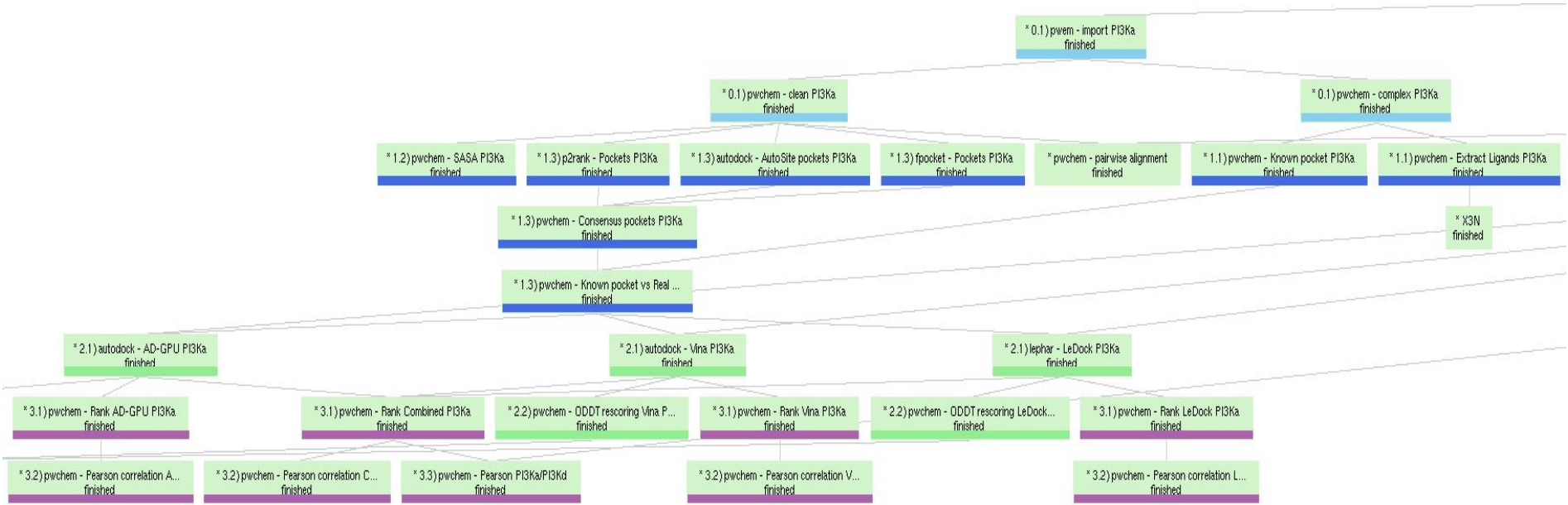
- **Virtual Drug Screening (VDS)**: workflow to enrich active molecules for a specific receptor.
- Progressively **heavier analysis** for **smaller amount of candidate molecules**
- **Selective:** able to catch the differences for active molecules in **receptor isoforms**



Selective VDS: workflow



Selective VDS: Scipion GUI

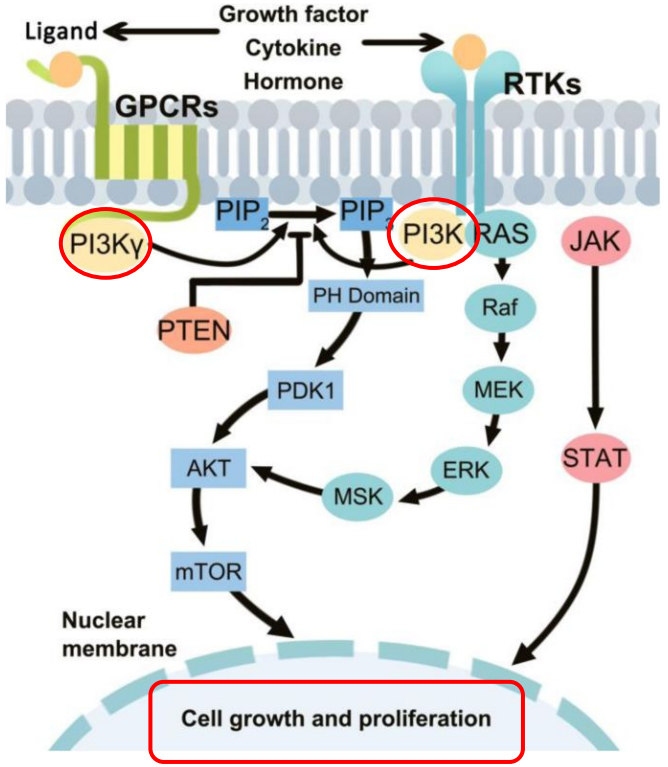


Selective VDS: case study PI3K

- **Receptor isoforms**: PI3K **α** and **δ**
- **PI3Ks** (Phosphatidyl Inositol 3-kinases)
- Receptor family that **phosphorylates** membrane lipids when activated
- Initiates **signal transduction cascades**

- Excessive **activation** ➡➡ Cancer
- Big amount of inhibitors but…

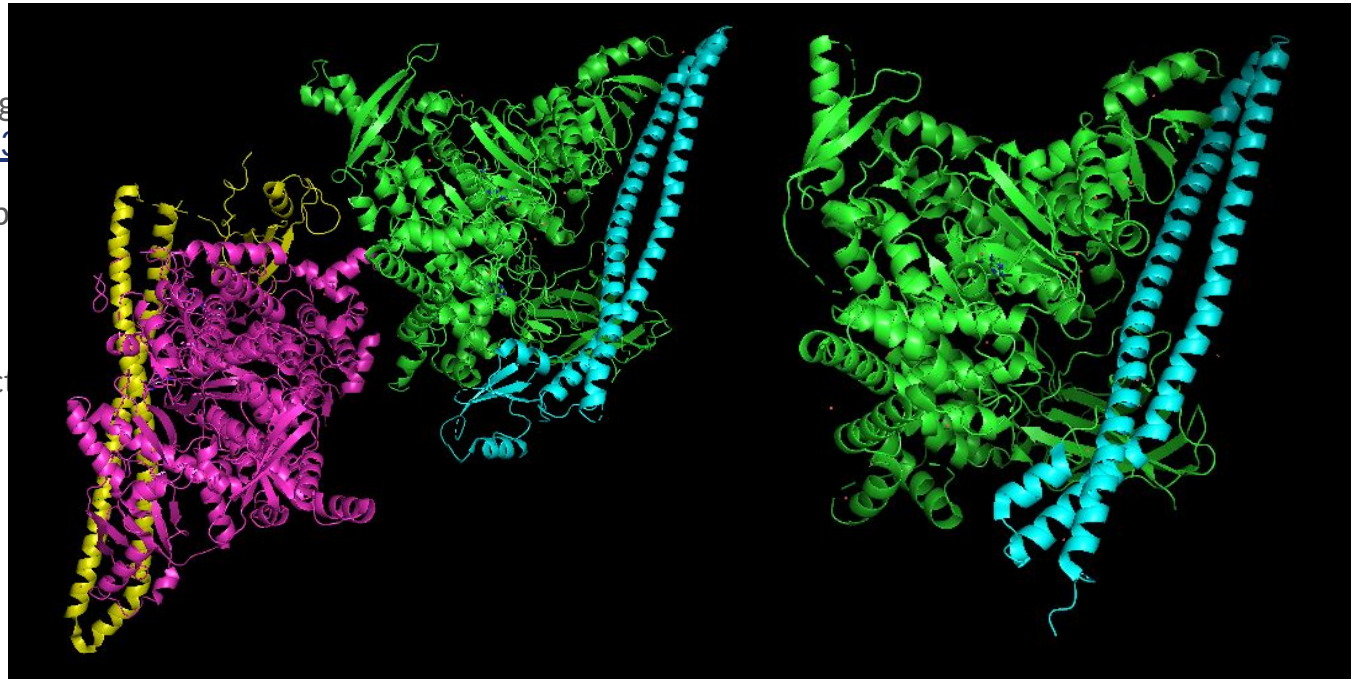
- **Non-specific inhibition** ➡➡ Side effects



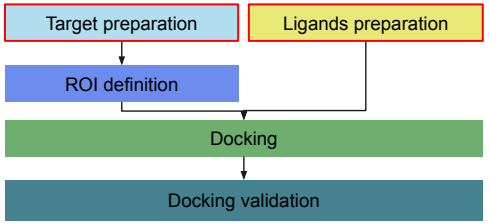
Selective VDS: experimental data

- Review of selective lig
  https://doi.org/10.103
- **13 ligands** with notab
  and **δ PI3K** isoforms

- Input PDB PI3K struct
  - **8TDU (α)**
  - **8BCY (δ)**



0) Imports and preparations



0.1) Receptor import and preparation:

- **Import** PDB **8TDU** (α), **8BCY** (δ)
- **Clean** structure
- Keep only **chain A**

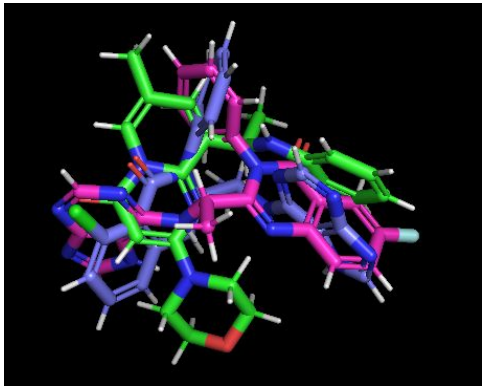
0.2) Complex study:

- **Import** PDB **8TDU** (α), **8BCY** (δ)
- Keep HETATMS (original ligands)
- Keep only **chain A**

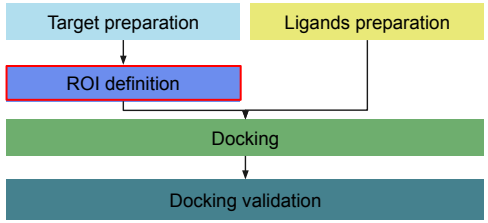
0.3) Ligand import and preparation:

- Downloading from **PubChem / local**
- Generate **3D structures**
- Generate **conformers**



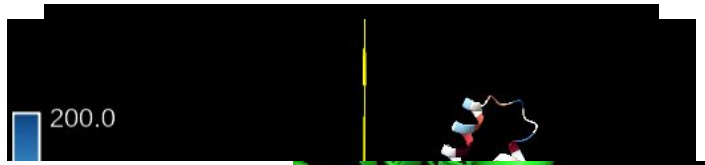


1) Druggable pocket analysis



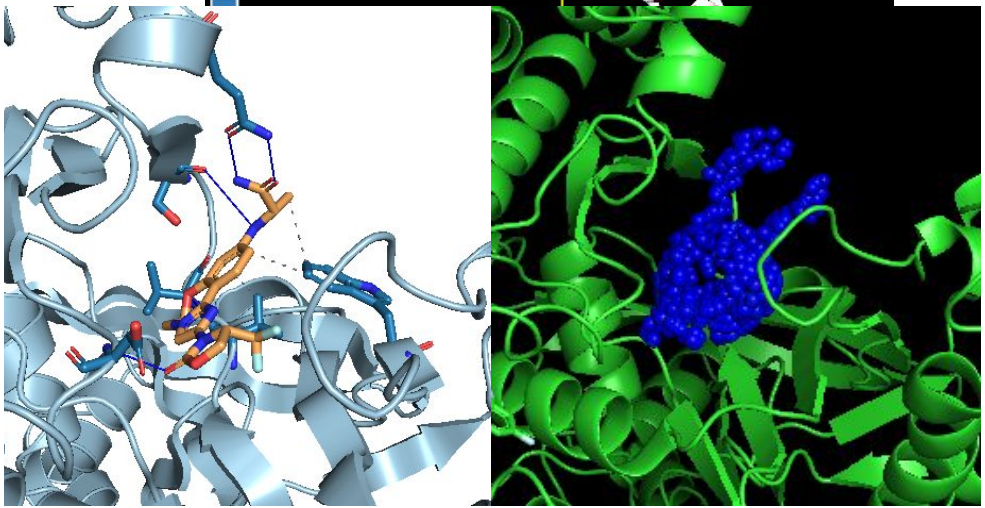
1.1) Pocket residues and contacts from complex ligands

- **Ligand extraction:** visualize interactions
- **Pocket definition**: comparison with predictions



1.2) **Receptor and pocket attribute analysis**

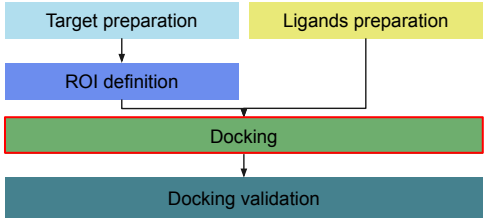
- **Surface accessibility** of the receptor



1.3) **Pocket prediction**

- Using **P2Rank, FPocket** and **AutoSite**
- **Consensus pockets**
- **Comparison** to real pocket -> Found!

2) Docking analysis

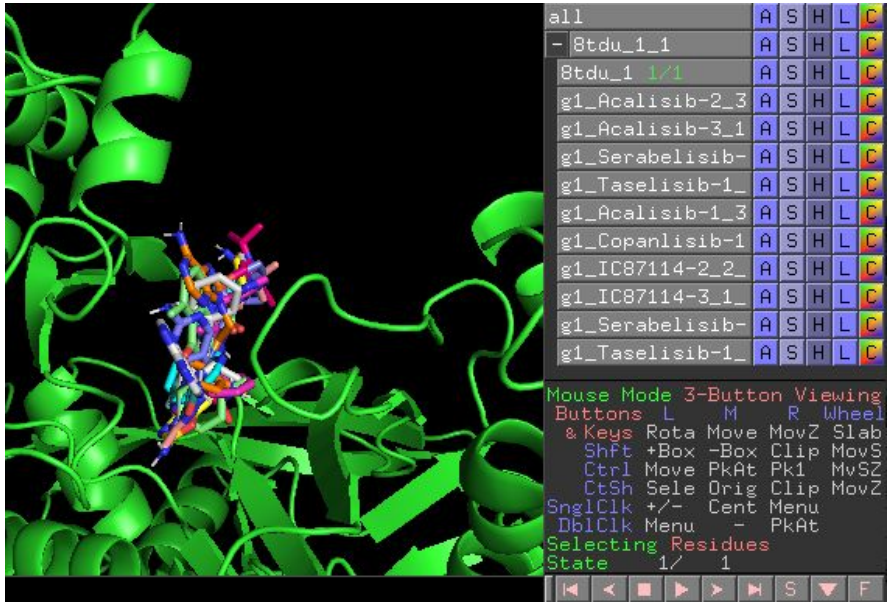


2.1) Docking over known pockets

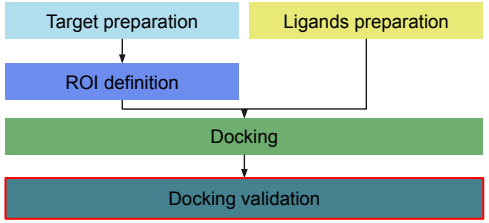
- Using **AutoDock-GPU, Vina** and **LeDock**
- Optionally, we can **filter the best results** (20%)

2.2) **Rescoring and consensus docking**

- **Rescore** using **ODDT** for comparability
- **Consensus** to keep **repeated poses**



3) Docking validation



3.1) Ranking score

- Algorithm to **aggregate docking results**
- **Normalized aggregation** of ranking (and scores) for each molecule in each docking run
- One molecule ➡➡ One score

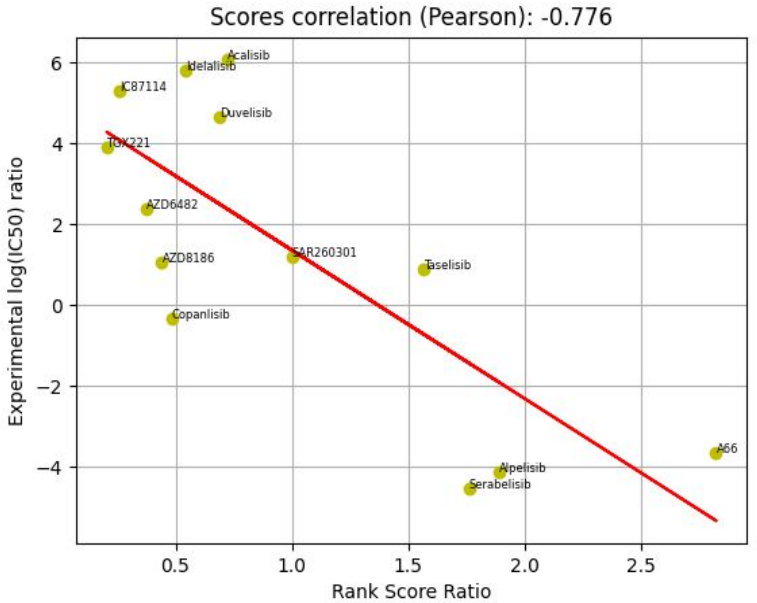
RankScore $_{(mol)}$ = $\sum_{(docks)}$ ($\sum_{(poses)}$ (1 - molRank/nMols)))

3.2) **Correlation** analysis (**VDS** validation):

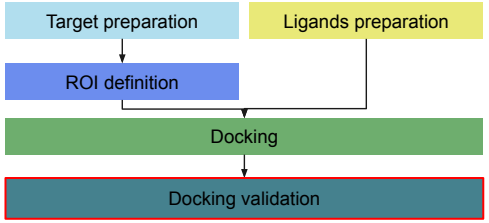
- Pearson correlation of **rank scores** to **experimental IC50** values (log)

3.3) **Correlation** analysis (**Selective VDS** validation):

- **Correlation** of rank scores and experimental IC50 ratios (log **α** / **δ**)

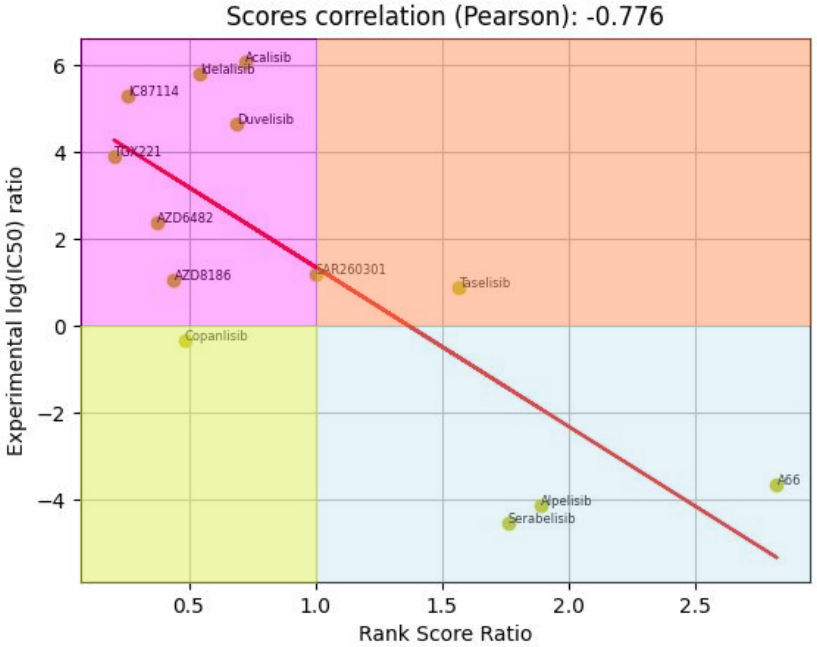


3) Docking validation



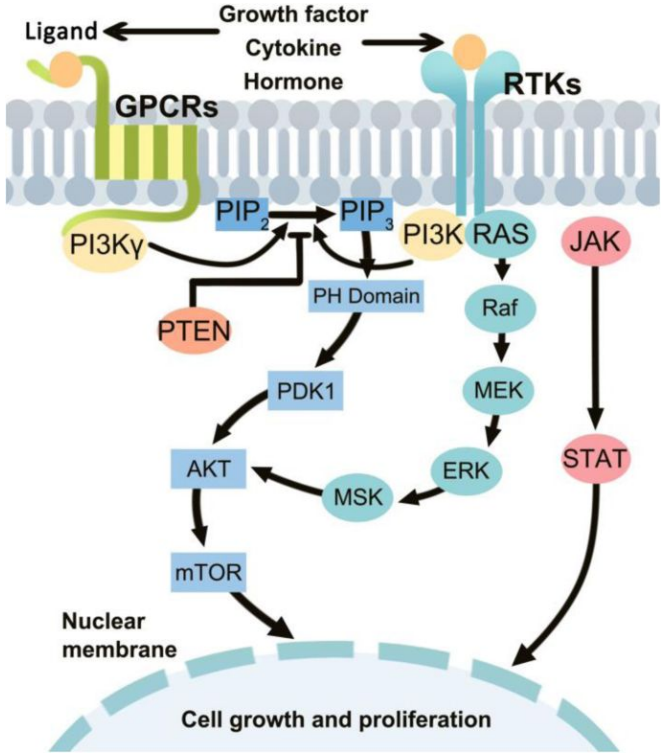
3.3) **Selective VDS** validation **(ratio α / δ)**

Name	PI3Kα	PI3Kδ
A66	32	1250
Acalisib	5441	12.7
Alpelisib	4.6	290
AZD6482	870	80
AZD8186	35	12
Copanlisib	0.5	0.7
Duvelisib	25	0.243
IC87114	>100000	500
Idelalisib	820	2.5
SAR260301	1539	468
Serabelisib	15	1390
Taselisib	0.29	0.12
TGX221	5000	100



Selective VDS: conclusions

- **Regular VDS** workflow further tested
- Tool for docking score aggregation: **rank scoring**
- Promising results for **selective VDS** in a real example
- Scipion-chem includes all the VDS tools and more:
  - **Lead optimization**
  - **Induced fit docking**
  - **Molecular dynamics** simulation



Selective VDS: play around

Play around in the selective VDS Scipion project or create a new project for **your receptor of interest**

Proposed activity A:

1) **Duplicate a protocol** to rerun a protocol without deleting the current results
2) **Perform a docking run** with a protocol that have not been used
3) **Visualize the interactions** of the molecule pose with best score for a docking run
4) **Filter** the input ligands using **ADME**
5) **Prepare a structure** with chains A and B from PI3Ka
6) Create a subset with the **best 5 docking poses** for AutoDock-GPU in PI3Ka

Proposed activity tips:

1) **Duplicate:** right click>Duplicate // Ctrl + D
2) Check **Vina-GPU**
3) Check the best scoring molecule using **table view** and open it with **PLIP**
4) Check **ADME** filter protocol
5) Using the **same protocol** used for obtaining the clean and complex structures
6) Use **“Filter set”** protocol

Selective VDS: play around

Play around in the selective VDS Scipion project or create a new project for **your receptor of interest**

Proposed activity B:

1) In PI3Ka, define a **pocket manually** for residues L1013, L938, L911, K941, Y1021, E950. Where are they located?
2) Launch a **docking on this pocket** and compare the energy results with those obtained from the active pocket.
3) Perform the **correlation analysis** on these new results.

Proposed activity tips:

1) Use the same protocol that defined the actual pocket using the existing ligands in the structure
2) Open both results (from the same software!) with the table view and sort by energy
3) Execute the correlation analysis with a duplicate of the correlation protocols and change the input to your docking

Selective VDS: play around

Play around in the selective VDS Scipion project or create a new project for **your receptor of interest**

Proposed activity C:

1) **Visualize the PI3Ka ligand X3N** in the original structure. Can you find a ligand in our list with a similar structure?
2) **Visualize and compare the docking poses** of both molecules. Can you determine which is the most probable pose of the question molecule?
3) What are **binding differences with X3N**?
4) Can you compare their **predicted interaction energies**?

Proposed activity tips:

1) Open in PyMol X3N as a molecule and compare it to the imported molecules
2) Open in Pymol X3N docked and compare them to the docking positions of the question molecule
3) Open X3N and the question molecule pose form question 2 using PLIP.
4) X3N has no energy since it does not originate from a docking. But rescoring protocols should be able to give an interacting energy prediction.